

A period-eight Chebyshev factorization for the zeros of a Jacobi-type pencil

Mathematical discovery run `fa_banach_001`

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Status. Candidate full solution, independently rechecked and likely valid, pending expert review.

1 Source problem

For the basic Jacobi-type pencil Θ_1 , Zagorodnyuk defines degree- n polynomials p_n and proves the representation

$$q_n(t) := p_n(\sqrt{2}t - 1) = T_n(t) + tU_{n-1}(t) - \frac{1}{2} \frac{U_{n-1}(t) - U_{n-1}(-1/\sqrt{2})}{t + 1/\sqrt{2}}, \quad (1)$$

initially away from $t = -1/\sqrt{2}$ and hence everywhere by polynomial continuation. Here T_j, U_j are the Chebyshev polynomials of the first and second kind and $U_{-1} = 0$. The source asks for the zero distribution and conjectures that all zeros are real [1, formula (8) and PDF p. 6].

By (13) and the last equalities we get

$$\begin{aligned} p_n(\sqrt{2}t - 1) &= \frac{1}{\sqrt{2}t + 1} \sin\left(\frac{3\pi}{4}n\right) + T_n(t) + \\ &+ \left(\frac{t}{\sqrt{1-t^2}i} - \frac{1}{(\sqrt{2}t + 1)\sqrt{2}\sqrt{1-t^2}i}\right) i \sin(n \arccos t), \end{aligned}$$

where $n \in \mathbb{Z}_+$, $t \in \left(-1, -\frac{1}{\sqrt{2}}\right) \cup \left(-\frac{1}{\sqrt{2}}, 1\right)$. Therefore, formula (8) holds. $\square$

By formula (8) or by the recurrent relation (6) one can calculate that

$$p_2(\lambda) = 2\lambda(\lambda + 2), \quad p_3(\lambda) = \sqrt{2}\lambda(2\lambda^2 + 6\lambda + 3).$$

We see that two subsequent polynomials have a common root 0. Thus, these polynomials are not orthogonal on the real line.

It is an interesting open problem to describe the distribution of zeros for polynomials p_n from (8). For example, we can conjecture that all zeros of p_n are real.

2 Idea of the proof

The divided difference in (1) hides a periodic factorization. Multiplication by $2(t + 1/\sqrt{2})$ turns q_n into $U_{n+1} + \sqrt{2}U_n + c_n$, where $c_n = \sqrt{2}\sin(3\pi n/4)$ depends only on n modulo eight. Elementary sum-to-product identities then factor this numerator into Chebyshev factors and fixed real combinations of consecutive Chebyshev polynomials. Such a combination has only simple real zeros by strict interlacing. The known linear factor $t+1/\sqrt{2}$ can therefore be removed without losing real-rootedness. Endpoint signs locate the unique exceptional zero below -1 .

3 Factorization and real-rootedness

Put

$$a = -\frac{1}{\sqrt{2}}, \quad R_j(t) = U_j(t) + \sqrt{2}U_{j-1}(t), \quad S_j(t) = T_j(t) + \sqrt{2}T_{j-1}(t).$$

Lemma 1. *For every $j \geq 1$, both R_j and S_j have j distinct real zeros. All zeros lie below 1. The polynomial R_j has one zero below -1 exactly when $j \geq 3$, whereas S_j always has exactly one zero below -1 . All remaining zeros lie in $(-1, 1)$.*

Proof. The zeros of consecutive members of either Chebyshev family are simple and strictly interlace. More generally, if P_j, P_{j-1} are consecutive members of a real orthogonal-polynomial family with positive leading coefficients, then $P_j + cP_{j-1}$ has one zero in every component of $\mathbb{R} \setminus Z(P_{j-1})$: at each zero of P_{j-1} its sign is the sign of P_j , and those signs alternate, including against the signs at $\pm\infty$. This supplies j distinct real roots and exhausts the degree. Apply this with $c = \sqrt{2}$ to U_j, U_{j-1} and T_j, T_{j-1} .

At 1, all four Chebyshev values occurring here are positive, so the largest root is below 1. At the other endpoint,

$$R_j(-1) = (-1)^j(j+1-\sqrt{2}j), \quad S_j(-1) = (-1)^j(1-\sqrt{2}).$$

The sign at $-\infty$ is $(-1)^j$. Thus the leftmost root of R_j crosses below -1 precisely when $j+1-\sqrt{2}j < 0$, i.e. $j \geq 3$; the leftmost root of S_j is always below -1 . $\square$

Theorem 2. *For every $n \geq 0$, all zeros of p_n are real and simple. More precisely, write $n = 2m - 1$ when n is odd and $n = 2m$ when it is even. Then*

$$2(t-a)q_n(t) = U_{n+1}(t) + \sqrt{2}U_n(t) + c_n, \quad c_n = \sqrt{2}\sin\frac{3\pi n}{4}, \quad (2)$$

and the numerator has the following factorization:

condition	c_n	$2(t-a)q_n(t)$
$n \equiv 0 \pmod{4}$	0	$R_{n+1}(t)$
$n = 2m - 1, c_n = 1$	1	$2T_m(t)R_m(t)$
$n = 2m - 1, c_n = -1$	-1	$2U_{m-1}(t)S_{m+1}(t)$
$n = 2m, c_n = -\sqrt{2}$	$-\sqrt{2}$	$2T_{m+1}(t)R_m(t)$
$n = 2m, c_n = \sqrt{2}$	$\sqrt{2}$	$2U_m(t)S_{m+1}(t)$

For $n = 1, 2, 3$, all n zeros of q_n lie in $(-1, 1)$. For every $n \geq 4$, exactly one zero lies in $(-\infty, -1)$ and the remaining $n - 1$ lie in $(-1, 1)$. Consequently, for $n \geq 4$, p_n has exactly one zero below $-1 - \sqrt{2}$ and all other zeros in $(-1 - \sqrt{2}, -1 + \sqrt{2})$. Finally, $\lambda = -2$ is a zero of p_n exactly when $n \equiv 2 \pmod{4}$.

Proof. The identity $T_n + tU_{n-1} = U_n$ and the recurrence $U_{n+1} = 2tU_n - U_{n-1}$ turn (1) into (2). Since $a = \cos(3\pi/4)$,

$$U_{n-1}(a) = \frac{\sin(3\pi n/4)}{\sin(3\pi/4)} = \sqrt{2} \sin \frac{3\pi n}{4};$$

the values for $n = 0, \dots, 7$ modulo eight are $0, 1, -\sqrt{2}, 1, 0, -1, \sqrt{2}, -1$.

The displayed factorization table follows from the standard product identities

$$\begin{aligned} 2T_m U_m &= U_{2m} + 1, & 2T_m U_{m-1} &= U_{2m-1}, \\ 2U_{m-1} T_{m+1} &= U_{2m} - 1, & 2U_{m-1} T_m &= U_{2m-1}, \\ 2T_{m+1} U_m &= U_{2m+1}, & 2T_{m+1} U_{m-1} &= U_{2m} - 1, \\ 2U_m T_{m+1} &= U_{2m+1}, & 2U_m T_m &= U_{2m} + 1. \end{aligned}$$

Every factor is real-rooted by Lemma 1 and the usual Chebyshev zero formulas. The numerator always vanishes at a , so division by $t - a$ proves real-rootedness of q_n .

It remains to check multiplicities. A zero common to T_m and R_m , or to U_{m-1} and S_{m+1} , would force $\cos \theta = -\sqrt{2}$, which is impossible. In the two even rows, the same trigonometric calculation shows that a common zero can only be $a = \cos(3\pi/4)$; it occurs exactly when $m \equiv 1$ or $3 \pmod{4}$, respectively, equivalently $n \equiv 2 \pmod{4}$. Thus the numerator has a double zero at a in precisely that congruence class, and division leaves one simple zero there. When $n \equiv 0 \pmod{4}$, Lemma 1 says the root a of R_{n+1} is simple and division removes it. All other numerator roots are simple and belong to only one factor. This proves simplicity and the final criterion $\lambda = \sqrt{2}a - 1 = -2$.

Finally, Lemma 1 shows that the R or S factor has exactly one root below -1 in every case with $n \geq 4$, while every ordinary T or U factor has its roots in $(-1, 1)$. The removable root a lies in $(-1, 1)$. Directly, q_1, q_2, q_3 have all roots in $(-1, 1)$. The affine map $\lambda = \sqrt{2}t - 1$ gives the stated intervals for p_n . $\square$

4 Verification and novelty bounds

The accompanying script checks the table by exact symbolic arithmetic for $0 \leq n \leq 80$ and checks real-rootedness, simplicity, and endpoint counts for the consecutive combinations through degree 200. The command

```
conda run --no-capture-output -n sandbox python
code/verify_factorizations.py
```

returns a pass for 81 exact factorizations and for the real-root/location checks through degree 200. These are regression checks; the proof above is exact.

A bounded novelty search on 21 August 2026 covered the four run indexes, exact arXiv-id and title searches, the source question's exact zero-distribution phrases, and later primary work on Jacobi-type pencils that cites the source. No later paper stating this factorization or resolving this exact zero problem was located. Novelty therefore remains subject to expert literature review.

Human review recommendation. Check the five factorization rows, the common-zero calculation at $a = -1/\sqrt{2}$, and the endpoint sign count in Lemma 1.

References

- [1] S. M. Zagorodnyuk, *Orthogonal polynomials related to some Jacobi-type pencils*, Ukrainian Math. J. **68** (2016), 1353–1365; arXiv:1508.01794.
- [2] G. Szegő, *Orthogonal Polynomials*, 4th ed., American Mathematical Society, 1975.